LECTURE 02

Story Proofs and Axioms of Probability

Typeset from the handwritten notes, with explanations and examples clarified using the lecture transcript.

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1 Counting: the key distinctions

Before counting, identify (i) whether an object can be chosen repeatedly and (ii) whether the order of the selections matters. For n available objects and k selections, the four standard cases are (with $k \le n$ when sampling without replacement):

	Order matters	**Order does not matter**
With replacement	n^k	$\binom{n+k-1}{k}$
Without replacement	$\dfrac{n!}{(n-k)!}$	$\binom{n}{k}$

Modeling advice from the lecture. Label people and objects when they represent distinct individuals (e.g., people $1, \dots, n$ or robberies $1, \dots, 6$). Count outcomes as equally likely *only when the chosen sample space justifies it.* To check an answer, test simple or extreme cases instead of merely rereading the calculation.

Example: splitting 10 people into teams. One team of 4 and another of 6 can be formed in $\binom{10}{4} = \binom{10}{6}$ ways: choosing one team determines the other. For *two unnamed teams of 5*, each division is counted twice when we choose the first group of 5, so the answer is

$$\frac{1}{2}\binom{10}{5}.$$

If the two teams have distinct labels (e.g., team A and team B), no division by 2 is needed.

2 Sampling with replacement when order does not matter

Question. We select k times from a set of n objects, returning the chosen object each time. If the order of selection does not matter, how many outcomes are possible?

Sampling with replacement, unordered: $\boxed{\binom{n+k-1}{k} = \binom{n+k-1}{n-1}}$

Here $n \ge 1$ and $k \ge 0$. Repeated selections are allowed; the result records only *how many times* each object was selected.

2.1 Check the formula in simple and extreme cases

1. $k = 0$**.** There is exactly one way to choose nothing:

$$\binom{n-1}{0} = 1.$$

   This agrees with the conventions $0! = 1$ and $\binom{n}{0} = 1$.

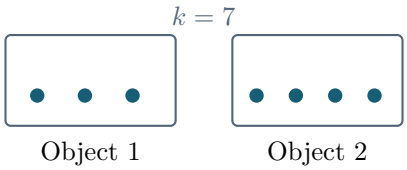
2. $k = 1$**.** We choose one of the n objects:

$$\binom{n}{1} = n.$$

3. $n = 2$**.** The proposed formula gives

$$\binom{k+1}{k} = \binom{k+1}{1} = k + 1.$$

   Indeed, if object 1 was selected j times, object 2 must have been selected $k - j$ times, where $j \in \{0, 1, \ldots, k\}$. There are exactly $k + 1$ possible values of j.

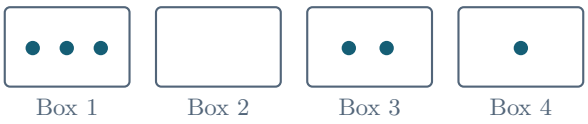


Example: 3 selections of object 1 and 4 selections of object 2.

2.2 Equivalent problem: indistinguishable particles and distinguishable boxes

The selection problem is equivalent to distributing k *indistinguishable particles* among n *distinguishable boxes*. Box i records how often object i was selected. The boxes are labeled $1, \ldots, n$, but only particle counts matter.

For example, with $n = 4$ boxes and $k = 6$ particles, consider the counts $(3, 0, 2, 1)$:



Four distinguishable boxes contain 3, 0, 2, and 1 indistinguishable particles.

2.3 Stars and bars proof

Represent each particle by a *star* ($\bullet$) and place a vertical bar ($|$) between neighboring boxes. The preceding example becomes

$$\underbrace{\bullet\bullet\bullet}_{\text{box 1}} \,|\, | \,\underbrace{\bullet\bullet}_{\text{box 3}} \,|\, \underbrace{\bullet}_{\text{box 4}}.$$

The two adjacent bars indicate that box 2 is empty.

Every allocation corresponds to *one and only one* sequence consisting of

- k stars, one for each particle;
- $n-1$ bars, to separate the n boxes.

There are $n+k-1$ positions in total. Choose which k positions contain stars; the remaining positions automatically contain bars. Thus,

$$\boxed{\text{Number of allocations} = \binom{n+k-1}{k}}.$$

Equivalently, choosing the $n-1$ bar positions yields

$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}.$$

Example: $n=4$ and $k=6$ give $\binom{9}{6} = \binom{9}{3} = 84$ possible allocations.

2.4 Clarification: coin flips are not equally likely "stars and bars" outcomes

The lecture connects indistinguishable-particle counting to the work of **Satyendra Nath Bose** and **Albert Einstein**. This is a contrast between *counting configurations* and *assigning probabilities*.

For two independent flips of a fair coin, the labeled outcomes are

$$\{HH, HT, TH, TT\}, \qquad P(HH) = P(HT) = P(TH) = P(TT) = \tfrac{1}{4}.$$

If we ignore order, these four outcomes fall into three categories:

$$\{HH\}, \quad \{HT, TH\}, \quad \{TT\}.$$

But the three categories are **not equally likely**:

$$\boxed{P(2H) = \tfrac{1}{4}, \qquad P(1H, 1T) = \tfrac{1}{2}, \qquad P(2T) = \tfrac{1}{4}.}$$

So the stars-and-bars formula can correctly *count* three unordered categories while giving no reason to assign each category probability 1/3. Bose's quantum-particle model uses different physical assumptions; it is *not* the standard probability model for ordinary fair coin tosses.

3 Story proofs: proving identities by interpretation

A **story proof** (or combinatorial proof) establishes an identity by *counting the same collection of objects in two different ways*. It proves the equation without requiring a long algebraic simplification.

3.1 Identity 1: binomial symmetry

$$\boxed{\binom{n}{k} = \binom{n}{n-k}}.$$

Story proof. To select a group of k from n people, we may choose who *belongs* to the group, or we may choose the $n-k$ people who are *left out*. Every choice in one description uniquely determines a choice in the other; therefore the counts agree.

3.2 Identity 2: choosing a committee and its president

$$\boxed{n\binom{n-1}{k-1} = k\binom{n}{k}.}$$

Story. From n people, form a committee of k members with one member designated as president.

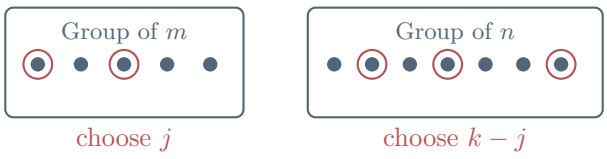
1. **President first.** Choose the president in n ways, then choose the remaining $k-1$ committee members from the other $n-1$ people:
$$n\binom{n-1}{k-1}.$$
2. **Committee first.** Choose the committee in $\binom{n}{k}$ ways, then choose its president from the k members:
$$k\binom{n}{k}.$$

Both procedures produce precisely the same committee-president pairs, proving the identity. (Assume $1 \le k \le n$.)

3.3 Identity 3: Vandermonde's identity

$$\boxed{\binom{m+n}{k} = \sum_{j=0}^{k} \binom{m}{j}\binom{n}{k-j}.}$$

Story. There are two disjoint groups: one with m people and another with n people. We want to choose k people from their union.



The circles mark the selected people. Exactly k are selected overall.

- **Count directly:** choose k people from all $m+n$ people in $\binom{m+n}{k}$ ways.
- **Count by the first group's contribution:** if exactly j come from the group of m, then exactly $k-j$ must come from the group of n. This gives $\binom{m}{j}\binom{n}{k-j}$ choices for that particular j.
- **Add disjoint cases:** sum over $j = 0, 1, \ldots, k$; no selection is counted twice, because every selection has a unique value of j.

This proves Vandermonde's identity. Use the convention $\binom{a}{b} = 0$ when $b < 0$ or $b > a$ for nonnegative integer a.

4 The non-naive definition of probability

The *naive* definition of probability counts favorable outcomes divided by all outcomes *only when the elementary outcomes are equally likely.* We now define probability without requiring equally likely outcomes or a finite sample space.

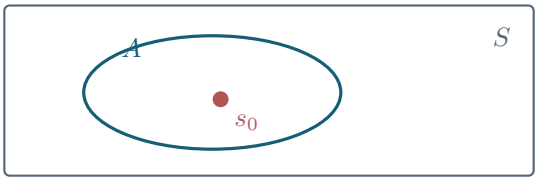
4.1 Probability space and events

The lecture writes a probability space as (S, P), where

- S is the **sample space**: the set of all possible outcomes of the experiment;
- an **event** A is a subset of S, written $A \subseteq S$;
- P is a **probability function**: it maps events to numbers between 0 and 1,

$$P : \mathcal{P}(S) \longrightarrow [0, 1], \qquad A \longmapsto P(A),$$

where $\mathcal{P}(S)$ denotes the collection of all subsets of S.



An event A occurs if the observed outcome s_0 belongs to A.

4.2 The two axioms given in the lecture

Axiom 1: impossible and certain events.

$$\boxed{P(\varnothing) = 0 \qquad \text{and} \qquad P(S) = 1.}$$

The empty event $\varnothing$ cannot occur; the full sample space S necessarily contains the observed outcome.

Axiom 2: countable additivity. If $A_1, A_2, \ldots$ are **pairwise disjoint** events, meaning $A_i \cap A_j = \varnothing$ whenever $i \neq j$, then

$$\boxed{P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).}$$

This states that the probability of a union of mutually non-overlapping events equals the sum of their probabilities, even for countably infinitely many events.

Mathematical precision. The lecture's notation $P : \mathcal{P}(S) \to [0, 1]$ is convenient at this stage. In the fully general theory, one usually specifies a *σ-algebra* $\mathcal{F} \subseteq \mathcal{P}(S)$ of allowable events and writes a probability space as $(S, \mathcal{F}, P)$, with $P : \mathcal{F} \to [0, 1]$. For this lecture, the two stated rules above are the main ideas.